Mettu Srinika

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Professional Summary

Computer Science (AI & ML) graduate with international research experience at Curtin University and hands-on experience building AI-powered applications, REST API-driven systems, and production-ready software. Developed OMIQORA, an AI-powered services ecosystem, applying backend development, system design, machine learning, and technical documentation to solve real-world engineering challenges.

Experience

- **Curtin University, Miri, Sarawak, Malaysia** — Miri, Malaysia
  Adversarial Machine Learning Research Intern — *Feb 2025 – April 2025*
  - **Research & Analysis**: Conducted research on adversarial machine learning in healthcare and finance by implementing and evaluating attack and defense techniques on real-world datasets.
  - **Implementation**: Implemented adversarial techniques including FGSM, PGD, One-Pixel, DeepFool, Adversarial Patch, and Model Inversion using TensorFlow.
  - **System Evaluation**: Evaluated model robustness using accuracy, precision, recall, F1-score, and adversarial testing methodologies.
  - **Documentation**: Prepared structured technical documentation covering methodology, experiments, findings, and reproducible results.

Projects

- **OMIQORA – AI-Powered Services Ecosystem** (Live Demo) — Personal Project
  Full-Stack AI Application
  - **System Design**: Designed a multi-role AI-powered services platform supporting customer, vendor, and admin workflows.
  - **Backend Development**: Built REST API-based backend features including authentication, booking, vendor onboarding, and payment workflows.
  - **Deployment**: Deployed the application on Vercel and validated end-to-end functionality across key user workflows.
  - **AI-Assisted Development**: Applied AI-assisted development workflows for implementation, debugging, testing, and technical documentation.
- **Adversarial Machine Learning in Healthcare & Finance**: Designed adversarial evaluation pipelines using TensorFlow and Scikit-learn; analyzed model vulnerabilities and improved post-attack accuracy through defensive strategies.
- **Smart Medical Report Analysis (LLMs + MCP)**: Developed automated medical report extraction and summarization using LLMs; implemented structured parsing and insight generation workflows.
- **Email Spam Classification (Naive Bayes vs. Decision Tree)**: Built and compared spam classification models using Naive Bayes and Decision Tree with TF-IDF feature extraction; achieved strong classification performance after preprocessing optimization. github.com/mettusrinika/EmailSpamHam

Achievements & Activities

- **SEED Business Challenge 2026**: Participated in an international business challenge by SEED Global Education with University of Maryland's Robert H. Smith School of Business, gaining exposure to global business strategy and entrepreneurship.
- **ISRO Bharatiya Antariksh Hackathon**: Participated in a national-level space technology hackathon.
- **IoT & Robotics Course**: Completed certified Skill Intern training program.
- **Volunteering**: Active volunteer in technical and cultural college programs.

Technical Skills

- **Programming**: Python, Java, JavaScript, HTML, CSS
- **AI & Machine Learning**: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, CNNs, Adversarial ML, LLM Systems
- **Backend & Databases**: REST APIs, MongoDB, MySQL
- **Tools**: Git, GitHub, Postman, VS Code, Jupyter Notebook

Education

- **Gokaraju Rangaraju Institute of Engineering and Technology** — Hyderabad, India
  B.Tech in Computer Science (AI & ML); CGPA: 8.83/10 — *2023 – 2026*

Certifications

- **Certified Hyperledger Developer**: NASSCOM
- **Python Programming**: Programming Hub
- **Introduction to Cybersecurity**: Cisco
- **MongoDB & Atlas Training**: MongoDB University
- **Data Structures Certification**: Programming Hub